DBMS: Data Design and Normalization

ASSIGNMENT 2.2

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STACKPULSE

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Outline

1 MULTIPLE CHOICE QUESTIONS (MCQs)

MCQ Q15: Database Normalization – Maximum Normal Form

Question MCQ Q15

Question: Consider the relation given below and find the maximum normal form applicable to them

1. R(A,B) with productions $\{A \rightarrow B\}$
2. R(A,B) with productions $\{B \rightarrow A\}$
3. R(A,B) with productions $\{A \rightarrow B, B \rightarrow A\}$
4. R(A,B,C) with productions $\{A \rightarrow B, B \rightarrow A, AB \rightarrow C\}$

Options:

(a) 1,2 and 3 are in 3NF and 4 is in BCNF

(b) 1 and 2 are in BCNF and 3 and 4 are in 3NF

(c) All are in 3NF

(d) All are in BCNF

MCQ Q15: Database Normalization – Strategy

Problem-Solving Strategy

- Find candidate keys for each relation
- Check if all determinants are candidate keys (BCNF condition)
- Binary relations are always in BCNF
- Check if any determinant is not a superkey

MCQ Q15: Database Normalization – Solution

Solution

Answer: (d) All are in BCNF

For R(A,B) with A→B: Candidate key = A → BCNF

For R(A,B) with B→A: Candidate key = B → BCNF

For R(A,B) with A→B, B→A: Candidate keys = A,B → BCNF

For R(A,B,C) with A→B, B→A, AB→C: Candidate keys = A, B. All determinants (A, B, AB) are superkeys → BCNF

MCQ Q16: Database Normalization – Highest Normal Form

Question MCQ Q16

Question: Consider the relation R(A,B,C,D,E) and the set F = $\{AB \rightarrow CE, E \rightarrow AB, C \rightarrow D\}$. What is the highest normal form of this relation?

Options:

(a) 1NF

(b) 2NF

(c) 3NF

(d) BCNF

MCQ Q16: Database Normalization – Strategy

Problem-Solving Strategy

- Find all candidate keys
- Identify prime and non-prime attributes
- Check for partial dependencies (2NF violation)
- Check for transitive dependencies (3NF violation)
- Check if all determinants are superkeys (BCNF condition)

MCQ Q16: Database Normalization – Solution

Solution

Answer: (b) 2NF

Candidate keys: AB and E (since AB→CE and E→AB)

Prime attributes: A, B, E

Non-prime attributes: C, D

Check 2NF: No partial dependencies as keys are single attributes or all attributes in key are prime

Check 3NF: C→D exists where C is non-prime and D is non-prime → transitive dependency exists

Therefore highest normal form is 2NF

MCQ Q17: Database Decomposition – Lossless or Lossy

Question MCQ Q17

Question: Given the relation R(X,Y,W,Z,P,Q) and the set F = $\{XY \to W, XW \to P, PQ \to Z, XY \to Q\}$. Consider the decomposition $R_1(Z, P, Q)$ and $R_2(X, Y, W, P, Q)$. This decomposition is:

Options:

(a) Lossless decomposition

(b) Lossy decomposition

(c) Either lossless or lossy

(d) Neither lossless or lossy

MCQ Q17: Database Decomposition – Strategy

Problem-Solving Strategy

- Find intersection of R1 and R2
- Check if intersection is a superkey for either relation
- Apply lossless join decomposition test: If R1 R2 $\rightarrow$ R1 or R1 R2 $\rightarrow$ R2, then decomposition is lossless

MCQ Q17: Database Decomposition – Solution

Solution

Answer: (a) Lossless decomposition

R1 R2 = {P, Q}

Check if {P,Q} is a key for R1: From PQ→Z, {P,Q} determines Z, so {P,Q} is a key for R1

Since intersection is a key for R1, decomposition is lossless

MCQ Q18: SCHEDULE Relation – Highest Normal Form

Question MCQ Q18

Question: Consider the relation SCHEDULE shown below. What is the highest normal form of this relation?
SCHEDULE (Student_ID, Class_No, Student_Name, Student_Major, Class_Time, Building_Room, Instructor).
Assume the following functional dependencies: {Student_ID $\rightarrow$ Student_Name, Student_ID $\rightarrow$ Student_Major, Class_No $\rightarrow$ Class_time, Class_No $\rightarrow$ Building_Room, Class_No $\rightarrow$ Instructor}

Options:

(a) 1NF

(b) 2NF

(c) 3NF

(d) BCNF

MCQ Q18: SCHEDULE Relation – Strategy

Problem-Solving Strategy

- Determine candidate key(s)
- Check for partial dependencies (dependencies on part of composite key)
- If partial dependencies exist, relation is not in 2NF, so highest is 1NF

MCQ Q18: SCHEDULE Relation – Solution

Solution

Answer: (a) 1NF

Candidate key: {Student_ID, Class_No} (composite key)

Partial dependencies exist:

- Student_ID→Student_Name (depends on part of key)
- Student_ID→Student_Major (depends on part of key)
- Class_No→Class_time (depends on part of key)
- Class_No→Building_Room (depends on part of key)
- Class_No→Instructor (depends on part of key)

Due to partial dependencies, relation is not in 2NF, so highest is 1NF

MCQ Q19: Lossless Decomposition

Question MCQ Q19

Question: A relation R is defined as R(S#, STATUS, CITY, SNAME) where S# is the primary key. If R is decomposed into two relations R1 and R2, which of the following is a lossless decomposition?

Options:

(a) R1(S#, STATUS), R2(S#, CITY, SNAME)

(b) R1(S#, STATUS), R2(STATUS, CITY, SNAME)

(c) R1(S#, STATUS, CITY), R2(CITY, SNAME)

(d) R1(S#, STATUS, SNAME), R2(CITY, STATUS)

MCQ Q19: Lossless Decomposition – Strategy

Problem-Solving Strategy

- For lossless decomposition, intersection of decomposed relations must be a superkey for at least one relation
- Check each option by finding R1 R2
- Verify if the intersection can determine all attributes of one relation

MCQ Q19: Lossless Decomposition – Solution

Solution

Answer: (a) R1(S#, STATUS), R2(S#, CITY, SNAME)

Option (a): R1 R2 = {S#} which is primary key in R1 → Lossless

Option (b): R1 R2 = {STATUS} not a key → Lossy

Option (c): R1 R2 = {CITY} not a key → Lossy

Option (d): R1 R2 = {STATUS} not a key → Lossy

MCQ Q20: Decomposition with Empty Intersection

Question MCQ Q20

Question: Consider the relation R(S, T, U, V) and the set F = $\{S \to T, T \to U, U \to V$ and $V \to S\}$. Let R is decomposed into R1 and R2 such that $R_1 \cap R_2 = \phi$, the decomposition is:

Options:

(a) Lossless decomposition

(b) Lossy decomposition

(c) Dependency preservation

(d) Cannot say

MCQ Q20: Decomposition with Empty Intersection – Strategy

Problem-Solving Strategy

- For lossless decomposition, intersection must be non-empty and a superkey
- Empty intersection means no common attributes to join on
- Without common attributes, original relation cannot be reconstructed

MCQ Q20: Decomposition with Empty Intersection – Solution

Solution

Answer: (b) Lossy decomposition

R1 R2 = means no common attributes

Without common attributes, we cannot perform natural join to reconstruct original relation

Therefore decomposition is lossy

MCQ Q21: Lossless Decomposition Test

Question MCQ Q21

Question: Suppose that we decompose the schema R(A, B, C, D, E) into R1(A, B, C) and R2(A, D, E) if the set F are functional dependencies holds on R. F $= \{A \rightarrow BC, CD \rightarrow E, B \rightarrow D, E \rightarrow A\}$. The decomposition is a:

Options:

(a) Lossless decomposition

(b) Lossy decomposition

(c) May or may not be lossless decomposition

(d) None of these

MCQ Q21: Lossless Decomposition Test – Strategy

Problem-Solving Strategy

- Find intersection of R1 and R2
- Check if intersection is a superkey for either relation
- Compute closure of intersection using given FDs

MCQ Q21: Lossless Decomposition Test – Solution

Solution

Answer: (a) Lossless decomposition

R1 R2 = {A}

Check if {A} is a key: From A→BC, we get B and C. From B→D, we get D. From CD→E, with C and D we get E. So A+ = {A,B,C,D,E}

Therefore A is a candidate key

Since intersection {A} is a key for R1, decomposition is lossless

MCQ Q22: Highest Normal Form Determination

Question MCQ Q22

Question: Find the highest normal form of the relation R(A, B, C, D) and the set F $= \{AB \rightarrow D, AC \rightarrow BD, B \rightarrow C\}$

Options:

(a) 1NF

(b) 2NF

(c) 3NF

(d) BCNF

MCQ Q22: Highest Normal Form Determination – Strategy

Problem-Solving Strategy

- Find all candidate keys
- Identify prime attributes
- Check for partial dependencies (2NF)
- Check for transitive dependencies (3NF)
- Check if all determinants are superkeys (BCNF)

MCQ Q22: Highest Normal Form Determination – Solution

Solution

Answer: (c) 3NF

Candidate keys: AB and AC

Prime attributes: {A, B, C}

Check 2NF: No partial dependencies as keys have A which is prime in all FDs

Check 3NF: All determinants (AB, AC, B) have:

- AB and AC are superkeys
- B→C: B is prime attribute (since B is in candidate key AB) and C is prime

Therefore relation is in 3NF

Not in BCNF because B→C has determinant B which is not a superkey

MCQ Q23: Inference of Functional Dependencies

Question MCQ Q23

Question: Consider the following set of functional dependencies (FDs): {Emp_No → Ename, Bdate, Address, Dept_No} {Dept_No → Dname, Mgr_No}. The additional FD(s) which cannot be inferred from the above set of FDs is:

Options:

(a) {Emp_No → Dname, Mgr_No}

(b) {Ename → Dept_No}

(c) {Emp_No → Dname}

(d) {Ename, Dept_No → Mgr_No}

MCQ Q23: Inference of Functional Dependencies – Strategy

Problem-Solving Strategy

- Compute closure of left-hand sides of each option
- Check if right-hand side attributes are in the closure
- Use given FDs to derive new FDs

MCQ Q23: Inference of Functional Dependencies – Solution

Solution

Answer: (b) {Ename → Dept_No}

{Emp_No}+ = {Ename, Bdate, Address, Dept_No, Dname, Mgr_No}

So Emp_No→Dname and Emp_No→Dname, Mgr_No can be inferred

{Ename, Dept_No}+ = {Ename, Dname, Mgr_No} (Dept_No→Dname, Mgr_No)

So Ename, Dept_No→Mgr_No can be inferred

{Ename}+ = {Ename} only

So Ename→Dept_No cannot be inferred

MCQ Q24: Properties of Relations

Question MCQ Q24

Question: A relation (from the relational database model) consists of a set of tuples, which implies that:

Options:

(a) all tuples in a relation must be distinct

(b) relational model supports multi-valued attributes whose values can be represented in sets

(c) for any two tuples, the values associated with all of their attributes may be the same

(d) all tuples in a particular relation may have different attributes

MCQ Q24: Properties of Relations – Strategy

Problem-Solving Strategy

- Recall fundamental properties of relations in relational model
- A relation is a set of tuples
- Sets, by definition, have no duplicate elements

MCQ Q24: Properties of Relations – Solution

Solution

Answer: (a) all tuples in a relation must be distinct

A relation is defined as a set of tuples

Sets, by mathematical definition, contain distinct elements (no duplicates)

Therefore all tuples in a relation must be distinct

MCQ Q25: Foreign Key Properties

Question MCQ Q25

Question: If K is a foreign key in relation R1, then:

Options:

(a) Every tuple of R1 has a distinct value for K

(b) K cannot have a null value for tuples in R1

(c) K is a key for some other relation

(d) K is a primary key for R1

MCQ Q25: Foreign Key Properties – Strategy

Problem-Solving Strategy

- Recall definition of foreign key
- Foreign key references primary key in another relation
- Foreign key can have duplicates and nulls (unless specified otherwise)

MCQ Q25: Foreign Key Properties – Solution

Solution

Answer: (c) K is a key for some other relation

Foreign key: An attribute that references a primary key in another relation

Option (c) correctly states: K is a key (primary key) for some other relation

Note: Foreign keys can have null values and duplicates in the referencing relation, so (a) and (b) are incorrect

MCQ Q26: Inference Rules for Functional Dependencies

Question MCQ Q26

Question: If W, X, Y and Z are attributes of a relation, which of the following inference rules for functional dependencies is correct?

Options:

(a) If $Y \subset X$ then $Y \rightarrow X$

(b) If $X \rightarrow Y$ then $(X, Z) \rightarrow (Y, W)$

(c) If $X \rightarrow (Y, Z)$ then $X \rightarrow Y$

(d) If $(X, Z) \rightarrow Y$ then $X \rightarrow Y$ and $Z \rightarrow Y$

MCQ Q26: Inference Rules – Strategy

Problem-Solving Strategy

- Recall Armstrong's axioms (reflexivity, augmentation, transitivity)
- Recall derived rules (decomposition, union, pseudotransitivity)
- Test each option against these rules

MCQ Q26: Inference Rules – Solution

Solution

Answer: (c) If $X \to (Y, Z)$ then $X \to Y$

Option (a): If YX then Y→X is false (subset doesn't determine superset) - violates reflexivity

Option (b): If X→Y then (X,Z)→(Y,W) is false (can't add W on right without reason)

Option (c): If X→(Y,Z) then X→Y is true (decomposition rule)

Option (d): If (X,Z)→Y then X→Y and Z→Y is false (union doesn't imply individual determinants)

MCQ Q27: BCNF Violation in Decomposition

Question MCQ Q27

Question: Consider relation schema R(A, B, C, D, E) with functional dependencies F containing $A \to B$ and $C \to D$. The schema is decomposed into $R_1(A, B)$ and $R_2(A, C, D, E)$. Find which relation is not in BCNF.

Options:

(a) R_1 is not in BCNF

(b) R_2 is not in BCNF

(c) R_1 is not in BCNF but R_2 is in BCNF

(d) None of these

MCQ Q27: BCNF Violation – Strategy

Problem-Solving Strategy

- Find FDs that hold in each decomposed relation
- Identify candidate keys for each relation
- Check BCNF condition: every determinant must be a superkey

MCQ Q27: BCNF Violation – Solution

Solution

Answer: (b) R_2 is not in BCNF

R1(A,B): Only FD A→B holds. A is a key for R1 → R1 is in BCNF

R2(A,C,D,E): FDs that hold: C→D (from original)

In R2, C is not a superkey (key of R2 would be {A,C}? A and C don't determine each other)

Since C→D has determinant C which is not a superkey in R2, R2 is not in BCNF